

Proof of the “irrelevance of the null SNPs” of the SER-relax model

1 Two helpful lemmas

Lemma 1. *Let R be a $J \times J$ positive semi-definite matrix, and let $x \in \mathbb{R}^J$ be a vector partitioned as $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, where x_1 is a scalar. Let S_1 be the Schur complement of the top-left scalar element R_{11} in R , and let S be a $J \times J$ matrix formed by padding S_1 with zeros such that the first row and column are zero, and the bottom-right $(J-1) \times (J-1)$ block is S_1 . For any positive scalar ζ , the following relationship holds:*

$$x^T(R + \zeta S)^{-1}x = \frac{1}{1 + \zeta} (x^T R^{-1}x) + \frac{\zeta}{1 + \zeta} (x_1^2 R_{11}^{-1})$$

Proof. Partition R into block form as $R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}$. The Schur complement of R_{11} is $S_1 = R_{22} - R_{21}R_{11}^{-1}R_{12}$.

Let $\tilde{R} = R + \zeta S$. Written in block form, this is:

$$\tilde{R} = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} + \zeta S_1 \end{pmatrix}$$

The Schur complement of R_{11} in $\tilde{R}$ is given by:

$$\tilde{S}_1 = (R_{22} + \zeta S_1) - R_{21}R_{11}^{-1}R_{12} = S_1 + \zeta S_1 = (1 + \zeta)S_1$$

Using the standard block matrix inversion formula, the inverse of R can be written as:

$$R^{-1} = \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + K S_1^{-1} K^T$$

where $K = \begin{pmatrix} -R_{11}^{-1}R_{12} \\ I \end{pmatrix}$.

Applying the same inversion formula to $\tilde{R}$, and noting that K remains unchanged, we obtain:

$$\tilde{R}^{-1} = \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + K \tilde{S}_1^{-1} K^T = \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{1 + \zeta} K S_1^{-1} K^T$$

From the decomposition of R^{-1} , we know that $K S_1^{-1} K^T = R^{-1} - \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix}$. Substituting this into the equation for $\tilde{R}^{-1}$ yields:

$$\tilde{R}^{-1} = \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{1 + \zeta} \left(R^{-1} - \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} \right)$$

$$\tilde{R}^{-1} = \frac{1}{1+\zeta} R^{-1} + \frac{\zeta}{1+\zeta} \begin{pmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix}$$

Finally, multiplying by x^T on the left and x on the right isolates $x_1^2 R_{11}^{-1}$ in the second term, resulting in:

$$x^T (R + \zeta S)^{-1} x = \frac{1}{1+\zeta} (x^T R^{-1} x) + \frac{\zeta}{1+\zeta} (x_1^2 R_{11}^{-1})$$

which completes the proof. $\square$

Lemma 2. *Let R , S , and ζ be defined as in Lemma 1. The determinant of the perturbed matrix $R + \zeta S$ satisfies the following relationship:*

$$\det(R + \zeta S) = (1 + \zeta)^{J-1} \det(R)$$

Proof. Using the determinant formula for block matrices, the determinant of R can be expressed as the product of the determinant of its top-left block and the determinant of its corresponding Schur complement:

$$\det(R) = \det(R_{11}) \det(S_1)$$

Applying the same block determinant formula to the perturbed matrix $\tilde{R} = R + \zeta S$, and recalling from Lemma 1 that its Schur complement is $\tilde{S}_1 = (1 + \zeta) S_1$, we have:

$$\det(R + \zeta S) = \det(R_{11}) \det((1 + \zeta) S_1)$$

Because S_1 is a matrix of dimension $(J-1) \times (J-1)$, factoring out the scalar $(1 + \zeta)$ from the determinant yields:

$$\det((1 + \zeta) S_1) = (1 + \zeta)^{J-1} \det(S_1)$$

Substituting this scalar property back into the determinant equation gives:

$$\det(R + \zeta S) = (1 + \zeta)^{J-1} \det(R_{11}) \det(S_1)$$

Since $\det(R_{11}) \det(S_1) = \det(R)$, we obtain the final result:

$$\det(R + \zeta S) = (1 + \zeta)^{J-1} \det(R)$$

which completes the proof. $\square$

2 Proof of the property

Here we will show that the PIP from SER-relax does not depend on the off-diagonal parts of $\bar{R}$. Recall the SER-relax model:

$$\begin{cases} x \mid z_j, \beta, \gamma = j \sim \mathcal{N} \left(\beta u_j(z_j), \frac{\bar{R}}{N} \right) \\ z_j \sim \mathcal{N}(\mu_j, \tau^2 S_j) \end{cases}$$

where $\mu_j = \bar{R}_{-j,j} \in \mathbb{R}^{J-1}$, $S_j = \bar{R}_{-j,-j} - \mu_j \mu_j^T$ is the Schur complement of $\bar{R}_{jj}$, and $u_j(z_j)$ is the vector with 1 on the j -element and the rest is z_j .

Goal. We want to calculate $p(\gamma = j \mid x, \beta)$ and $p(\gamma = j \mid x)$, and show that it only depends on $\bar{R}$ through

$$q = x^T \bar{R}^{-1} x.$$

Calculation for $j = 1$.

$$\begin{cases} x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix} \mid z_1, \beta, \gamma = 1 \sim \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{\bar{R}}{N}\right) \\ z_1 \sim \mathcal{N}(\mu_1, \tau^2 S_1) \end{cases}$$

where $\mu_1 = \bar{R}_{-1,1}$, $S_1 = \bar{R}_{-1,-1} - \mu_1 \mu_1^T$.

Marginal & Conditional: Marginalize out z_1 :

$$x \mid \beta, \gamma = 1 \sim \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} 1 & \mu_1^T \\ \mu_1 & \bar{R}_{-1,-1} + N\beta^2 \tau^2 S_1 \end{pmatrix}\right)$$

Let

$$\tilde{R} = \begin{pmatrix} 1 & \mu_1^T \\ \mu_1 & \bar{R}_{-1,-1} + N\beta^2 \tau^2 S_1 \end{pmatrix} = \bar{R} + \begin{pmatrix} 0 & 0 \\ 0 & N\beta^2 \tau^2 S_1 \end{pmatrix}.$$

We have

$$p(x \mid \beta, \gamma = 1) = \frac{N^{J/2}}{(2\pi)^{J/2} |\tilde{R}|^{1/2}} \exp\left(-\frac{N}{2} \left(x - \beta \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}\right)^T \tilde{R}^{-1} \left(x - \beta \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}\right)\right)$$

1. Because $\begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}$ is the first column of $\tilde{R}$,

$$\tilde{R}^{-1} \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix} = e_1, \quad \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}^T \tilde{R}^{-1} \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix} = 1.$$

2. Using Lemma 1 with $\zeta = N\beta^2 \tau^2$ we have

$$x^T \tilde{R}^{-1} x = \frac{1}{1 + N\beta^2 \tau^2} x^T \bar{R}^{-1} x + \frac{N\beta^2 \tau^2}{1 + N\beta^2 \tau^2} x_1^2.$$

3. Using Lemma 2:

$$\det(\tilde{R}) = (1 + N\beta^2 \tau^2)^{J-1} \det(\bar{R}).$$

Combine all of them and let

$$q = x^T \bar{R}^{-1} x,$$

we have

$$\begin{aligned} p(x \mid \beta, \gamma = 1) &= \frac{N^{J/2}}{(2\pi)^{J/2} |\bar{R}|^{1/2} (1 + N\beta^2 \tau^2)^{(J-1)/2}} \exp\left(-\frac{N}{2(1 + N\beta^2 \tau^2)} q\right) \\ &\times \exp\left(-\frac{N^2 \beta^2 \tau^2}{2(1 + N\beta^2 \tau^2)} x_1^2\right) \exp\left(N\beta x_1 - \frac{N}{2} \beta^2\right). \end{aligned}$$

Similarly,

$$p(x \mid \beta, \gamma = j) = \frac{N^{J/2}}{(2\pi)^{J/2} |\bar{R}|^{1/2} (1 + N\beta^2\tau^2)^{(J-1)/2}} \exp\left(-\frac{N}{2(1 + N\beta^2\tau^2)}q\right) \\ \times \exp\left(-\frac{N^2\beta^2\tau^2}{2(1 + N\beta^2\tau^2)}x_j^2\right) \exp\left(N\beta x_j - \frac{N}{2}\beta^2\right). \quad (1)$$

Conclusion for $p(\gamma = j \mid x)$. After integrating over β , the q -dependent term remains inside the integral. Dropping constants that are common to all j ,

$$p(\gamma = j \mid x) \propto \int (1 + N\beta^2\tau^2)^{-(J-1)/2} \exp\left(-\frac{N}{2(1 + N\beta^2\tau^2)}q\right) \\ \times \exp\left(-\frac{N^2\beta^2\tau^2}{2(1 + N\beta^2\tau^2)}x_j^2\right) \times \exp\left(N\beta x_j - \frac{N}{2}\beta^2\right) p(\beta \mid 0, \sigma_0^2) d\beta.$$

Now combine the last exponential factor with the Gaussian prior on β . Let

$$s_N^2 = \frac{1}{N + \sigma_0^{-2}}, \quad m_j = \frac{N}{N + \sigma_0^{-2}}x_j,$$

so that $m_j \approx x_j$ and s_N^2 is small when N is large. Completing the square gives

$$\exp\left(N\beta x_j - \frac{N}{2}\beta^2\right) p(\beta \mid 0, \sigma_0^2) = C_j p(\beta \mid m_j, s_N^2),$$

where

$$C_j = \frac{1}{\sigma_0 \sqrt{N + \sigma_0^{-2}}} \exp\left(\frac{\sigma_0^2 N}{\sigma_0^2 N + 1} \frac{Nx_j^2}{2}\right).$$

Therefore the same expression can be written as

$$p(\gamma = j \mid x) \propto \mathbb{E}_{\beta \sim \mathcal{N}(m_j, s_N^2)} \left[(1 + N\beta^2\tau^2)^{-(J-1)/2} \exp\left(-\frac{N}{2(1 + N\beta^2\tau^2)}q\right) \right. \\ \left. \times \exp\left(\left(\frac{N\sigma_0^2}{1 + N\sigma_0^2} - \frac{N\beta^2\tau^2}{1 + N\beta^2\tau^2}\right) \frac{Nx_j^2}{2}\right) \right].$$

Implication for PIP diffuseness. This expectation form gives a useful way to compare SER-relax to SER-relax-prime when ν_0 is small. Define

$$z_j = \sqrt{N}x_j, \quad \rho = \frac{N\sigma_0^2}{1 + N\sigma_0^2}, \quad r(a) = \frac{a}{1 + a}.$$

It is useful to change variables to

$$U = \sqrt{N}\beta.$$

Since

$$\beta \mid x_j \sim \mathcal{N}(m_j, s_N^2), \quad m_j = \rho x_j, \quad s_N^2 = \frac{1}{N + \sigma_0^{-2}},$$

we have the exact induced distribution

$$U \mid x_j \sim \mathcal{N}(\rho z_j, \rho).$$

When $N\sigma_0^2$ is large, $\rho \approx 1$, so this is approximately

$$U \mid x_j \sim \mathcal{N}(z_j, 1).$$

Thus the original SER-relax expression can be written as

$$p(\gamma = j \mid x) \propto \mathbb{E}_{U \sim \mathcal{N}(\rho z_j, \rho)} \left[G(U^2) \exp \left\{ \frac{1}{2} (\rho - r(\tau^2 U^2)) z_j^2 \right\} \right],$$

where

$$G(U^2) = (1 + \tau^2 U^2)^{-(J-1)/2} \exp \left\{ -\frac{Nq}{2(1 + \tau^2 U^2)} \right\}.$$

This is the main distinction from SER-relax-prime. In the original model, the relaxation scale inside the integral is $\tau^2 U^2$, and the distribution of U is centered at the SNP-specific value ρz_j . Thus a SNP with a large z_j is allowed to put mass on large U^2 , whereas a null SNP with $z_j \approx 0$ has U centered near zero. To reach the same large U^2 as the top SNP, a null SNP must pay a Gaussian tail cost under $\mathcal{N}(\rho z_j, \rho)$.

A simple plug-in approximation replaces U by its mean ρz_j :

$$\log p(\gamma = j \mid x) \approx \frac{1}{2} \{ \rho - r(\tau^2 \rho^2 z_j^2) \} z_j^2 - \frac{J-1}{2} \log(1 + \tau^2 \rho^2 z_j^2) - \frac{Nq}{2(1 + \tau^2 \rho^2 z_j^2)} + \text{constant}.$$

When $\rho \approx 1$, this becomes the simpler approximation with $\tau^2 z_j^2$. This plug-in formula is only an approximation; the exact statement is the expectation over $U \sim \mathcal{N}(\rho z_j, \rho)$ above.

In SER-relax-prime, β^2 in the covariance is replaced by the fixed prior scale σ_0^2 . Consequently the determinant and q terms are common across j and cancel in the normalization over SNPs. The resulting single-SNP weight is

$$\tilde{p}(\gamma = j \mid x) \propto \exp \left\{ \frac{1}{2} [\rho - r(N\sigma_0^2 \tau^2)] z_j^2 \right\}.$$

Thus SER-relax-prime applies the same attenuation to the z_j^2 evidence for every SNP:

$$\text{SER-relax-prime attenuation: } r(N\sigma_0^2 \tau^2).$$

The exact log-weight gap between the top SNP $j_\star$ and another SNP k in SER-relax-prime is

$$\log \tilde{p}(\gamma = j_\star \mid x) - \log \tilde{p}(\gamma = k \mid x) = \frac{1}{2} \{ \rho - r(N\sigma_0^2 \tau^2) \} (z_{j_\star}^2 - z_k^2).$$

If σ_0^2 is learned and is of the order of the largest observed x_j^2 , then

$$N\sigma_0^2 \approx z_{j_\star}^2.$$

Therefore the prime attenuation can be similar to the original model's plug-in attenuation at the top SNP. This observation is important: the difference between the two models is not necessarily

that the top SNP is less attenuated in original SER-relax. The difference is that SER-relax-prime applies the top-SNP-scale attenuation globally to every SNP. If $N\sigma_0^2\tau^2$ is large, then

$$r(N\sigma_0^2\tau^2) \approx \rho \approx 1,$$

and the coefficient of every z_j^2 is close to zero. The log-weight gaps collapse, and the PIPs become diffuse.

In original SER-relax, the analogous relaxation is not global. It is integrated against

$$U \mid x_j \sim \mathcal{N}(\rho z_j, \rho).$$

For the top SNP, this distribution is centered near z_{j*} , so the integral can use large U^2 values and obtain LD-column flexibility. For a null SNP, the distribution is centered near zero, so the same large relaxation requires a tail event. Consequently null SNPs do not receive the same effective relaxation as the top SNP. This SNP-specific integration can preserve the ranking and keep the highest PIP on the SNP with the largest Z -score, even when the learned σ_0^2 is of the order of $\max_j x_j^2$ and ν_0 is small.

Taylor expansion around the prime scale. The preceding comparison can be made more explicit by expanding the original SER-relax expectation around the fixed scale used by SER-relax-prime. Let

$$a = N\sigma_0^2, \quad b = 1 + \tau^2 a, \quad S = U^2, \quad \Delta = S - a.$$

Write the log of the non-Gaussian part of the original expectation as

$$H_j(s) = -\frac{J-1}{2} \log(1 + \tau^2 s) - \frac{Nq}{2(1 + \tau^2 s)} + \frac{1}{2} \{\rho - r(\tau^2 s)\} z_j^2.$$

Then

$$p(\gamma = j \mid x) \propto \mathbb{E}_{U \sim \mathcal{N}(\rho z_j, \rho)} [\exp\{H_j(U^2)\}].$$

A Taylor expansion around $s = a$ gives

$$H_j(a + \Delta) = H_j(a) + A_j \Delta + \frac{1}{2} B_j \Delta^2 + O(\Delta^3),$$

where

$$H_j(a) = -\frac{J-1}{2} \log b - \frac{Nq}{2b} + \frac{1}{2} \{\rho - r(\tau^2 a)\} z_j^2,$$

$$A_j = \frac{\tau^2}{2} \left\{ \frac{Nq - z_j^2}{b^2} - \frac{J-1}{b} \right\},$$

and

$$B_j = \frac{\tau^4}{2} \left\{ \frac{J-1}{b^2} - \frac{2(Nq - z_j^2)}{b^3} \right\}.$$

The zeroth-order term is

$$\exp\{H_j(a)\} \propto \exp \left\{ \frac{1}{2} [\rho - r(N\sigma_0^2\tau^2)] z_j^2 \right\},$$

because the determinant and q terms in $H_j(a)$ do not depend on j . This is exactly the SER-relax-prime weight. Thus SER-relax-prime can be viewed as the zeroth-order approximation to original SER-relax obtained by setting

$$U^2 = N\beta^2 = N\sigma_0^2$$

for every SNP.

The first correction is SNP-specific because the distribution of $\Delta = U^2 - a$ depends on z_j . Under $U \sim \mathcal{N}(\rho z_j, \rho)$,

$$\mathbb{E}_j[\Delta] = \rho + \rho^2 z_j^2 - a,$$

and

$$\text{Var}_j(\Delta) = 2\rho^2 + 4\rho^3 z_j^2.$$

Keeping terms through second order gives the approximation

$$\log p(\gamma = j \mid x) \approx H_j(a) + A_j \mathbb{E}_j[\Delta] + \frac{1}{2} B_j \mathbb{E}_j[\Delta^2] + \frac{1}{2} A_j^2 \text{Var}_j(\Delta) + \text{constant}.$$

This expression shows where SER-relax can depart from SER-relax-prime. If $j_\star$ is the strongest SNP and the learned prior variance satisfies $a = N\sigma_0^2 \approx z_{j_\star}^2$, then

$$\mathbb{E}_{j_\star}[U^2] \approx a$$

up to an $O(1)$ correction when a is large. Therefore the prime zeroth-order approximation is locally reasonable for the top SNP. For a null SNP with $z_j \approx 0$, however,

$$\mathbb{E}_j[U^2] \approx \rho \approx 1, \quad \mathbb{E}_j[\Delta] \approx 1 - a,$$

so U^2 is not close to a at all. The Taylor expansion around a is then not a good local approximation for that SNP. This is precisely the difference: SER-relax-prime evaluates all SNPs at the same top-SNP-scale value $U^2 = a$, whereas original SER-relax evaluates each SNP by integrating over a distribution centered at its own ρz_j .

This also clarifies the role of the $G(U^2)$ term. At zeroth order, $G(a)$ is common across SNPs and cancels, which is what happens in SER-relax-prime. In original SER-relax, $G(U^2)$ remains inside the SNP-specific expectation. A strong SNP, whose U distribution is centered far from zero, can place mass on large U^2 values and access the corresponding LD-column flexibility. A null SNP, whose U distribution is centered near zero, cannot access the same part of $G(U^2)$ without paying a Gaussian tail cost. Therefore even if σ_0^2 is learned to be close to the largest x_j^2 , the original model does not give every SNP the same effective relaxation scale.

Suggested diagnostics. For each SNP, define the normalized SER-relax integrand in the U scale,

$$\pi_j(dU) \propto \phi(U; \rho z_j, \rho) G(U^2) \exp \left\{ \frac{1}{2} (\rho - r(\tau^2 U^2)) z_j^2 \right\} dU,$$

where $\phi(\cdot; \mu, v)$ is the $\mathcal{N}(\mu, v)$ density. A direct diagnostic is to compute

$$\mathbb{E}_{\pi_j}[\tau^2 U^2]$$

for each SNP and compare it with the prime value

$$N\sigma_0^2 \tau^2.$$

The expected pattern is that top SNPs have larger effective relaxation, null SNPs have much smaller effective relaxation, and SER-relax-prime uses one common relaxation scale for all SNPs. A second diagnostic is to compare the actual SER-relax PIPs with the plug-in approximation obtained by replacing U by ρz_j in the expectation above.

Therefore $p(\gamma = j \mid x)$ depends on $\bar{R}$ through $q = x^T \bar{R}^{-1} x$ in general. It does not depend on the off-diagonal elements of $\bar{R}$ separately once q is fixed.

2.1 Approximate irrelevance of the null SNPs

Another way to understand the behavior of SER-relax is to condition on the marginal statistic x_j . Write

$$x = \begin{pmatrix} x_j \\ x_{-j} \end{pmatrix}, \quad \bar{R} = \begin{pmatrix} 1 & \mu_j^T \\ \mu_j & \bar{R}_{-j,-j} \end{pmatrix}, \quad S_j = \bar{R}_{-j,-j} - \mu_j \mu_j^T.$$

In both susie-rss and SER-relax, the marginal distribution of x_j is

$$x_j \mid \beta, \gamma = j \sim \mathcal{N}(\beta, 1/N).$$

Therefore the posterior based only on x_j is

$$\beta \mid x_j, \gamma = j \sim \mathcal{N}(m_j, s_N^2), \quad m_j = \rho x_j, \quad s_N^2 = \frac{\rho}{N}, \quad \rho = \frac{N\sigma_0^2}{1 + N\sigma_0^2}.$$

In susie-rss, the conditional distribution of the remaining coordinates is independent of β :

$$p(x_{-j} \mid x_j, \beta, \gamma = j) = \mathcal{N}(x_j \mu_j, S_j/N).$$

This is the exact irrelevance property: once x_j is fixed, x_{-j} does not change the relative evidence for SNP j through β .

In SER-relax, the conditional mean is still independent of β , but the conditional covariance is inflated by β^2 :

$$p(x_{-j} \mid x_j, \beta, \gamma = j) = \mathcal{N}\left(x_j \mu_j, \left(\frac{1}{N} + \beta^2 \tau^2\right) S_j\right).$$

Integrating over $\beta \mid x_j$ gives

$$\begin{aligned} p(x_{-j} \mid x_j, \gamma = j) &= \int \mathcal{N}\left(x_{-j} \mid x_j \mu_j, \left(\frac{1}{N} + \beta^2 \tau^2\right) S_j\right) p(\beta \mid x_j, \gamma = j) d\beta \\ &= \mathbb{E}_{\beta \sim \mathcal{N}(m_j, s_N^2)} \left[\mathcal{N}\left(x_{-j} \mid x_j \mu_j, \left(\frac{1}{N} + \beta^2 \tau^2\right) S_j\right) \right]. \end{aligned}$$

This is a mixture of Gaussians with the same mean and the same shape matrix S_j , but with SNP-specific scalar variance inflation. Since $\beta \mid x_j$ is concentrated near $m_j = \rho x_j$, a plug-in approximation gives

$$p(x_{-j} \mid x_j, \gamma = j) \approx \mathcal{N}\left(x_j \mu_j, \left(\frac{1}{N} + m_j^2 \tau^2\right) S_j\right).$$

When $\rho \approx 1$, this becomes

$$p(x_{-j} \mid x_j, \gamma = j) \approx \mathcal{N}\left(x_j \mu_j, \left(\frac{1}{N} + x_j^2 \tau^2\right) S_j\right).$$

This gives an approximate null-SNP irrelevance statement. If SNP j is null, then $x_j = O(N^{-1/2})$, and under $\beta \mid x_j$,

$$\beta^2 = O_p(N^{-1}).$$

Therefore the scalar covariance inflation satisfies

$$\frac{1}{N} + \beta^2 \tau^2 = \frac{1}{N} \{1 + O_p(\tau^2)\}.$$

For moderate τ^2 , the SER-relax conditional distribution is close to the susie-rss conditional distribution:

$$p(x_{-j} \mid x_j, \gamma = j) \approx \mathcal{N}(x_j \mu_j, S_j/N).$$

In contrast, for a strong SNP with z_j^2 large, the inflation $1 + N\beta^2\tau^2$ can be much larger than 1 under $\beta \mid x_j, \gamma = j$, allowing the active SNP's LD column to relax. This is the approximate version of irrelevance: null SNPs see little variance inflation after conditioning on their small x_j , while strong SNPs can receive substantial LD-column flexibility.

This argument also shows why using $p(\beta \mid x, \gamma = j)$ directly is less clean. The full posterior $p(\beta \mid x, \gamma = j)$ already contains x_{-j} through the Gaussian likelihood above, so it is circular for proving irrelevance of x_{-j} . The sharper statement is obtained by first conditioning on x_j and then asking how much the remaining conditional distribution changes as β varies over $\beta \mid x_j, \gamma = j$.

3 TO-DO

Understand the relationship between

1. PIP only depends on x and the diagonal of R .
2. $p(x_{-j} \mid x_j, \beta, \gamma = j)$ does not depend on β .

Look Yuxin's thesis and read the SI of susie-rss paper more carefully.

Why does having β^2 in the variance make sense? Of course it follows from the IW or Wi model, but intuitively why does it make sense?

Is there another model that arises from the conditional distribution:

$$p(x_{-j} \mid x_j, \beta, \gamma = j) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0 + 3}\right) S_j\right)? \quad (2)$$

4 Some new models with better approximation to susie-love

Recall our original SER model from what is called 'susie-love':

$$\begin{cases} x \mid R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, R/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \quad (3)$$

Note that the latent matrix R appears in both mean and variance of the likelihood in (3). This model is very natural for unknown in-sample LD R , but is hard to make inferences. Last time we approximated R in the variance to be $\bar{R}$, which makes the inference much easier but might eventually lead to the loss of the "irrelevance of the null SNPs" property. In principle, it makes

sense that when $L = 1$, PIP of SNP j calculation should only depend on x_j but not on other x_k and $\bar{R}_{jk}$ (“irrelevance of the null SNPs” property for SER).

I think this original model is actually feasible for inference. Consider the following equivalent way to represent it.

4.1 Model A (equivalent to the original SER-love model)

Let’s write down $p(x|R, \beta, \gamma = 1)$, other cases of γ are similar. Let

$$d_1 = R_{11} \in \mathbb{R}_+, \quad z_1 = \frac{R_{-1,1}}{d_1} \in \mathbb{R}^{J-1},$$

and

$$S_1 = R_{-1,-1} - d_1 z_1 z_1^\top$$

be the Schur’s complement of R_{11} in R . Hence, the latent R is represented as

$$R = \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}. \quad (4)$$

Similarly, denote

$$\bar{S}_1 = \bar{R}_{-1,-1} - \bar{R}_{-1,1} \bar{R}_{-1,1}^\top$$

be the Schur’s complement of $\bar{R}_{11}$ in $\bar{R}$, and $\mu_1 = \bar{R}_{-1,1}$, we can write

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}. \quad (5)$$

Properties of IW distribution (together with the fact that $\bar{R}_{11} = 1$) imply, using the shape–scale parameterization for the inverse-gamma distribution,

$$\begin{cases} d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\bar{R}_{-1,1}, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \end{cases} \quad (6)$$

The original SER version of Susie-love (with $\gamma = 1$; similar for other γ) becomes:

$$\begin{cases} x | \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right) \\ d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (7)$$

Deriving $p(x \mid \gamma = 1)$. Write

$$x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}.$$

Step 1: likelihood conditional on all latent variables. From the block normal distribution in (7),

$$p(x \mid \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}_J \left(x \mid \beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix} \right).$$

Equivalently, using the conditional distribution of a block normal,

$$p(x \mid \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N} \left(x_1 \mid \beta d_1, \frac{d_1}{N} \right) \mathcal{N}_{J-1} \left(x_{-1} \mid x_1 z_1, \frac{S_1}{N} \right).$$

The key simplification is that, **after conditioning on x_1 , the conditional mean of x_{-1} is $x_1 z_1$ and no longer contains β or d_1** (relevant to our earlier discussion).

Step 2: marginalize over $z_1 \mid S_1$. Since

$$z_1 \mid S_1 \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{S_1}{\nu_0} \right)$$

and

$$x_{-1} \mid x_1, z_1, S_1 \sim \mathcal{N}_{J-1} \left(x_1 z_1, \frac{S_1}{N} \right),$$

we get

$$x_{-1} \mid x_1, S_1, \gamma = 1 \sim \mathcal{N}_{J-1} \left(x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) S_1 \right). \quad (8)$$

Therefore

$$p(x \mid \gamma = 1, \beta, d_1, S_1) = \mathcal{N} \left(x_1 \mid \beta d_1, \frac{d_1}{N} \right) \mathcal{N}_{J-1} \left(x_{-1} \mid x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) S_1 \right).$$

What I found so interesting is that the conditional distribution $p(x_{-1} \mid x_1, S_1, \gamma = 1)$ resembles what was suggested by Matthew last time (cf. (2)).

Step 3: marginalize over S_1 . Because $S_1 \sim \mathcal{IW}_{J-1}(\nu_0 \bar{S}_1, \nu_0 + J + 1)$, integrating S_1 gives a multivariate t distribution with $\nu_0 + 3$ degrees of freedom:

$$x_{-1} \mid x_1, \gamma = 1 \sim t_{J-1, \nu_0+3} \left(x_1 \mu_1, \frac{\nu_0}{\nu_0 + 3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right),$$

where the second argument is the multivariate t scale matrix. Thus

$$p(x \mid \gamma = 1, \beta, d_1) = \mathcal{N} \left(x_1 \mid \beta d_1, \frac{d_1}{N} \right) t_{J-1, \nu_0+3} \left(x_{-1} \mid x_1 \mu_1, \frac{\nu_0}{\nu_0 + 3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right).$$

Equivalently, the t factor is proportional to

$$|\bar{S}_1|^{-1/2} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right)^{-(J-1)/2} \left[1 + \frac{(x_{-1} - x_1\mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} \right]^{-(\nu_0+J+2)/2}.$$

Using the block inverse formula (notice the block form of $\bar{R}$ (5)),

$$x^\top \bar{R}^{-1} x = x_1^2 + (x_{-1} - x_1\mu_1)^\top \bar{S}_1^{-1} (x_{-1} - x_1\mu_1),$$

so

$$1 + \frac{(x_{-1} - x_1\mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} = \frac{x^\top \bar{R}^{-1} x + \nu_0/N}{x_1^2 + \nu_0/N}.$$

Therefore

$$T_1(x) \propto |\bar{S}_1|^{-1/2} \left(x_1^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N} \right)^{-(\nu_0+J+2)/2}. \quad (9)$$

When comparing different values of γ , the final factor is common across SNPs. Also, since $\bar{R}_{11} = 1$, $|\bar{R}| = |\bar{S}_1|$; analogously, $|\bar{S}_j| = |\bar{R}|$ for any SNP j with $\bar{R}_{jj} = 1$.

Step 4: marginalize over β and d_1 . The preceding display separates the problem into an $x_{-1} \mid x_1$ factor and an x_1 factor. Define

$$T_1(x) = t_{J-1, \nu_0+3} \left(x_{-1} \mid x_1\mu_1, \frac{\nu_0}{\nu_0+3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right).$$

If we integrate over β , then

$$x_1 \mid d_1, \gamma = 1 \sim \mathcal{N} \left(0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right),$$

and hence (finally integrating over d_1):

$$p(x \mid \gamma = 1) = T_1(x) \int_0^\infty \mathcal{N} \left(x_1 \mid 0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right) \text{IG} \left(d_1; \frac{\nu_0+2}{2}, \frac{\nu_0}{2} \right) dd_1.$$

This is a one-dimensional integral over $d_1 > 0$. I do not see a standard closed form for it because the variance is $d_1/N + d_1^2 \sigma_0^2$. Replacing T_1 with its expression in (9),

$$\begin{aligned} p(x \mid \gamma = 1) &\propto |\bar{R}|^{-1/2} \left(x_1^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N} \right)^{-(\nu_0+J+2)/2} \\ &\quad \times \int_0^\infty \mathcal{N} \left(x_1 \mid 0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right) d \text{IG}(d_1) \end{aligned}$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1} x$ do not depend on which SNP is chosen, both factors cancel in the PIP normalization. Also,

$$\left(x_j^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} = \left(\frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(1 + \frac{N x_j^2}{\nu_0} \right)^{(\nu_0+3)/2},$$

and the first factor is common across SNPs. Hence,

$$p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \int_0^\infty \mathcal{N}\left(x_j \mid 0, \frac{d_j}{N} + d_j^2 \sigma_0^2\right) d\text{IG}(d_j).$$

Conclusion. In the original SER-love model, the marginal likelihood depends on $x^\top \bar{R}^{-1}x$, but this dependence is common across SNPs and cancels when computing the PIP. The resulting PIP weight depends on SNP j only through x_j .

4.2 Model B

Simplify $d_1 \equiv 1$ to avoid intractable integral (and we know the diagonal of the in-sample LD matrix is 1 anyway), we have the following model (model B):

$$\begin{cases} x|\beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix}\right) \\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (10)$$

This model leads to

$$p(x \mid \gamma = 1) \propto |\bar{R}|^{-1/2} \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \mathcal{N}\left(x_1 \mid 0, \frac{1}{N} + \sigma_0^2\right).$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1}x$ do not depend on which SNP is chosen, the PIP weights simplify further. Dropping the common factors gives

$$p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right). \quad (11)$$

Compare to SER-RSS's PIP. Recall SER-RSS's PIP:

$$\tilde{p}(\gamma = j \mid x) \propto \exp\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right) \quad (12)$$

We now show that (11) tends to (12) as $\nu_0 \rightarrow \infty$. Since

$$\left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} = \left(1 + \frac{2}{\nu_0 + 3} \times \frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right)^{(\nu_0+3)/2} \asymp \exp\left\{\frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right\}$$

Also,

$$\mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right) \propto \exp\left\{-\frac{x_j^2}{2(1/N + \sigma_0^2)}\right\} = \exp\left\{-\frac{1}{1 + N\sigma_0^2} \frac{Nx_j^2}{2}\right\}.$$

Therefore the limiting weight in (11) is asymptotically (as ν_0 gets large)

$$\exp \left\{ \left(\frac{\nu_0 + 3}{\nu_0} - \frac{1}{N\sigma_0^2 + 1} \right) \frac{Nx_j^2}{2} \right\} = \exp \left\{ \left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} - \frac{1}{\nu_0 + 3} \right) \frac{Nx_j^2}{2} \right\} \approx \exp \left\{ \frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2} \right\}$$

which is exactly the SER-RSS PIP weight in (12). Hence, for finite ν_0 , PIP under SER-love model (B) is slightly more diffused than susie-rss.

Inference under model B (10). For a general SNP j , write

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \quad \kappa_j = \nu_0 + Nx_j^2.$$

Under model B, conditional on $\gamma = j$, the likelihood factorizes as

$$p(x \mid \beta, z_j, S_j, \gamma = j) = \mathcal{N} \left(x_j \mid \beta, \frac{1}{N} \right) \mathcal{N}_{J-1} \left(x_{-j} \mid x_j z_j, \frac{S_j}{N} \right).$$

Thus β is conditionally independent of (z_j, S_j) given x and $\gamma = j$, and

$$p(\beta, z_j, S_j \mid x, \gamma = j) = p(\beta \mid x_j, \gamma = j) p(z_j, S_j \mid x, \gamma = j).$$

- The posterior for β is the usual normal-normal update:

$$\boxed{\beta \mid x, \gamma = j \sim \mathcal{N} \left(\frac{N}{N + \sigma_0^{-2}} x_j, \frac{1}{N + \sigma_0^{-2}} \right).}$$

- For the LD-column variables, first condition on S_j . Combining

$$z_j \mid S_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{S_j}{\nu_0} \right), \quad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}_{J-1} \left(x_j z_j, \frac{S_j}{N} \right)$$

gives

$$\boxed{z_j \mid S_j, x, \gamma = j \sim \mathcal{N}_{J-1} \left(\frac{\nu_0 \mu_j + Nx_j x_{-j}}{\nu_0 + Nx_j^2}, \frac{S_j}{\nu_0 + Nx_j^2} \right).}$$

- Integrating out z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}_{J-1} \left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0} \right) S_j \right).$$

Therefore the posterior for S_j is inverse-Wishart:

$$\boxed{S_j \mid x, \gamma = j \sim \mathcal{IW} \left(\nu_0 \bar{S}_j + \frac{N\nu_0}{\nu_0 + Nx_j^2} (x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top, \nu_0 + J + 2 \right).}$$

Equivalently, the joint posterior of (z_j, S_j) can be represented by first drawing S_j from this inverse-Wishart distribution and then drawing z_j from the conditional normal distribution above. Finally, the full posterior over γ uses the PIP weights in (11), normalized over $j = 1, \dots, J$.

Closed-form ELBO. Because the conditional posteriors above are exact, the optimized ELBO for model B is the exact marginal log likelihood. Let

$$V_\beta = \frac{1}{N} + \sigma_0^2, \quad q = x^\top \bar{R}^{-1} x.$$

For a uniform prior on γ , the optimized ELBO is

$$\mathcal{L}(\nu_0, \sigma_0^2) = \log \left\{ \frac{1}{J} \sum_{j=1}^J \exp(\ell_j) \right\},$$

where the component log marginal likelihood can be written as

$$\begin{aligned} \ell_j = & -\frac{1}{2} \log(2\pi V_\beta) - \frac{x_j^2}{2V_\beta} + \log \Gamma\left(\frac{\nu_0 + J + 2}{2}\right) - \log \Gamma\left(\frac{\nu_0 + 3}{2}\right) \\ & - \frac{J-1}{2} \log \pi - \frac{1}{2} \log |\bar{S}_j| + \frac{\nu_0 + 3}{2} \log\left(x_j^2 + \frac{\nu_0}{N}\right) - \frac{\nu_0 + J + 2}{2} \log\left(q + \frac{\nu_0}{N}\right). \end{aligned}$$

This is equivalent to the marginal likelihood derived above, but unlike the PIP formula it keeps the terms that are common across SNPs. Those common terms do not affect $\pi_j = p(\gamma = j \mid x)$ for fixed hyperparameters, but they do matter when learning ν_0 .

Empirical Bayes learning of ν_0 and σ_0^2 . The empirical Bayes estimates can be obtained by maximizing the closed-form objective

$$(\hat{\nu}_0, \hat{\sigma}_0^2) = \arg \max_{\nu_0 > 0, \sigma_0^2 \geq 0} \mathcal{L}(\nu_0, \sigma_0^2).$$

After choosing $(\hat{\nu}_0, \hat{\sigma}_0^2)$, the posterior inclusion probabilities are

$$\pi_j = \frac{\exp(\ell_j)}{\sum_{k=1}^J \exp(\ell_k)} \Big|_{\nu_0 = \hat{\nu}_0, \sigma_0^2 = \hat{\sigma}_0^2}.$$

Equivalently, for computing PIPs at fixed hyperparameters one may use the simplified weights in (11); for empirical Bayes optimization one should use $\mathcal{L}(\nu_0, \sigma_0^2)$ above, because it retains the ν_0 -dependent normalizing constants and the common $\log(q + \nu_0/N)$ term.

EM view of empirical Bayes. Another way to use the fact that the posterior is exact is to write the usual ELBO identity. Let

$$\theta = (\gamma, \beta, z_\gamma, S_\gamma), \quad q(\theta) = p(\theta \mid x; \nu_0^{\text{old}}, \sigma_{0,\text{old}}^2).$$

Strictly speaking,

$$\log p(x; \nu_0, \sigma_0^2) \geq \mathbb{E}_q\{\log p(x \mid \theta)\} - \text{KL}\{q(\theta) \parallel p(\theta; \nu_0, \sigma_0^2)\},$$

with equality when q is the posterior under the same hyperparameters. In an EM algorithm, the E-step fixes q at the current hyperparameters and the M-step maximizes

$$Q(\nu_0, \sigma_0^2) = \mathbb{E}_q\{\log p(\theta; \nu_0, \sigma_0^2)\},$$

because the likelihood term does not involve ν_0 or σ_0^2 in model B.

This gives a closed-form update for σ_0^2 . If

$$\pi_j = q(\gamma = j), \quad m_{\beta j} = \frac{N}{N + \sigma_{0,\text{old}}^{-2}} x_j, \quad v_\beta = \frac{1}{N + \sigma_{0,\text{old}}^{-2}},$$

then

$$\sigma_{0,\text{new}}^2 = \sum_{j=1}^J \pi_j (m_{\beta j}^2 + v_\beta).$$

For ν_0 , the same EM view reduces the update to a one-dimensional optimization, but not to a closed-form expression. Let $p = J - 1$ and write the current posterior of S_j as

$$S_j \mid x, \gamma = j \sim \mathcal{IW}_p(\Psi_j, m_j), \quad m_j = \nu_0^{\text{old}} + J + 2,$$

where

$$\Psi_j = \nu_0^{\text{old}} \bar{S}_j + \frac{N \nu_0^{\text{old}}}{\nu_0^{\text{old}} + N x_j^2} (x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top.$$

Also let

$$\bar{z}_j = \frac{\nu_0^{\text{old}} \mu_j + N x_j x_{-j}}{\nu_0^{\text{old}} + N x_j^2}, \quad \kappa_j^{\text{old}} = \nu_0^{\text{old}} + N x_j^2.$$

The needed posterior expectations are

$$A_j = \mathbb{E}_q\{\text{tr}(\bar{S}_j S_j^{-1})\} = m_j \text{tr}(\bar{S}_j \Psi_j^{-1}),$$

$$B_j = \mathbb{E}_q\{(z_j - \mu_j)^\top S_j^{-1} (z_j - \mu_j)\} = m_j (\bar{z}_j - \mu_j)^\top \Psi_j^{-1} (\bar{z}_j - \mu_j) + \frac{p}{\kappa_j^{\text{old}}},$$

and

$$C_j = \mathbb{E}_q\{\log |S_j|\} = \log |\Psi_j| - \sum_{r=1}^p \psi\left(\frac{m_j + 1 - r}{2}\right) - p \log 2.$$

Up to constants that do not depend on ν_0 , the EM objective for ν_0 is

$$\begin{aligned} Q_\nu(\nu_0) = \sum_{j=1}^J \pi_j \left[\frac{p}{2} \log \nu_0 + \frac{\nu_0 + J + 1}{2} \log |\nu_0 \bar{S}_j| - \frac{(\nu_0 + J + 1)p}{2} \log 2 \right. \\ \left. - \log \Gamma_p\left(\frac{\nu_0 + J + 1}{2}\right) - \frac{\nu_0 + 2J + 2}{2} C_j - \frac{\nu_0}{2} (A_j + B_j) \right]. \end{aligned}$$

Thus

$$\nu_{0,\text{new}} = \arg \max_{\nu_0 > 0} Q_\nu(\nu_0).$$

So the exact-posterior ELBO gives a particularly nice update for σ_0^2 , and a stable scalar optimization for ν_0 . For ν_0 , the closed-form marginal likelihood $\mathcal{L}(\nu_0, \sigma_0^2)$ above is often even simpler to optimize directly, but the EM form is useful for deriving a monotone coordinate-ascent algorithm.

5 Extension to susie

Now we try to extend model B into a Susie-like model, where we allow more than one non-zero effect.